TALES OF THE OMEGA · PAPER T

The Torsion Condensate:

Saturation Bound Framework for Vacuum Stiffness

and Geometric Phase Discrimination

$\alpha_K = \theta/\blacksquare_{Pl}^2 \cdot G_v \sim 10^{15}$ Pa (LQG) · Biased defect channel · 11.967 h · 3.932 min/day kill-switch

D. Kadunic · Omega Research Collective / MSDS Programme · April 2026

Hybrid phenomenological model · Companion: Papers R, S, W · Target: Foundations of Physics

ABSTRACT

We present a saturation-bound parametrisation for the torsion condensate. The dimensionless coupling $\alpha_K = 1.7\times10^{-4}$ is interpreted as $\alpha_K = \theta/\blacksquare_{Pl}^2$, mapping the observed constant onto a geometric grain size $\sqrt{\theta} \sim 2\times10^{-37}$ m. Drawing on LQG area eigenvalues for dimensional estimates (hybrid phenomenological model), we estimate an effective Vacuum Shear Modulus $G_v \sim 10^{15}$ Pa as the energy cost of deforming a spin-network link against axial-vector currents, with geometric dilution factor $(\blacksquare_{Pl}/L)^2$ transducing Planck to nuclear scale. The condensate is described by a Ginzburg-Landau potential (mapped to Paper O), with α_K identified as the dimensionless reduced density $\varepsilon = (\rho_c - \rho)/\rho_c$. The sidereal phase drift of 3.932 min/day is the experiment's kill-switch discriminant. The biased defect channel is the paper's strongest pillar: the torsion field modulates an existing materials-science signal; the modulation depth is α_K, independent of the absolute baseline. All relations are testable hypotheses. The data will decide.

PAPER T: FIVE PILLARS OF THE SATURATION BOUND FRAMEWORK

1. $\alpha K = \theta/\blacksquare Pl^2$: maps coupling to geometric grain size $\sqrt{\theta} \sim 2\times10^{-37}$ m
2. $Gv \sim 10^{15}$ Pa: LQG area eigenvalue + nucleon Compton scale + $(lPl/L)^2$ dilution
3. Ginzburg-Landau: $\alpha K = \varepsilon$ (reduced density), MSDS condensate just below critical
4. Sidereal kill-switch: 3.932 min/day drift — galactic frame vs solar frame
5. Biased defect channel: modulate existing (200) signal — strongest pillar

1. Introduction: From Phenomenology to Saturation Bounds

Papers R and S established the experimental protocol and the heuristic 10^{-4} scale. Paper T supplies saturation bounds and effective parametrisations for the torsion condensate. We address four questions without claiming first-principles quantum-gravity derivation: (1) What is α_K? (2) Why does the vacuum have stiffness? (3) Why is $\alpha_K = 1.7\times10^{-4}$ the critical value? (4) Why does the 3.932 min/day sidereal drift discriminate against all systematics?

FRAMEWORK STATUS: Paper T is a hybrid phenomenological model. We draw on non-commutative geometry for the area bound, Loop Quantum Gravity for the energy cost, and Ginzburg-Landau theory for the phase transition. These are not claimed to be fundamentally unified; they are used for dimensional estimates that converge on the observed 10^{-4} scale. All relations are presented as testable hypotheses.

2. The Saturation Bound: $\alpha K = \theta/\blacksquare Pl^2$

2.1 Non-commutative geometry parametrisation

$$\alpha_K \equiv \theta / \blacksquare_{Pl}^2$$

θ: non-commutative area scale · $\blacksquare_{Pl} = 1.616\times10^{-35}$ m

$$\theta = \alpha_K \times \blacksquare_{Pl}^2 = 1.7\times10^{-4} \times (1.616\times10^{-35})^2 = 4.43\text{e-}74\ \text{m}^2$$

$\sqrt{\theta}$ = 2.11e-37 m · far below LHC bound (~10^{-18} m) — not contradicted

This is a parametrisation — we map the observed α_K onto a geometric grain size, not a derivation from a specific non-commutative model. Note: the earlier attempt to define a torsion correlation length $\xi = \blacksquare_{Pl}/\alpha_K \sim 10^{-31}$ m (smaller than $\blacksquare_{Pl}$) was physically inconsistent and has been removed.

3. Vacuum Shear Modulus from LQG Phenomenology

3.1 LQG minimum area eigenvalue

$$A_{min} = 8\pi\gamma\, \blacksquare_{Pl}^2 \times \sqrt{3}/2 = 1.348\text{e-}69\ \text{m}^2$$

γ = 0.2375 (Immirzi parameter) · j = 1/2 (fundamental link) · smallest possible area — the spacetime pixel

3.2 G_v with geometric dilution (step-by-step, verified)

$$G_v = \alpha_K \times (\blacksquare c/A_{min}) \times (1/L) \times (\blacksquare_{Pl}/L)^2$$

L ~ 1 fm = 10^{-15} m (nucleon Compton scale) · $(\blacksquare_{Pl}/L)^2$ = geometric dilution

Step	Calculation	Result
$\blacksquare c$	1.055e-34 J·s × 3.00e+08 m/s	3.165e-26 J·m
$\blacksquare c / A_{min}$	3.165e-26 / 1.348e-69 m^2	2.348e+43 J/m^2
$\times \alpha_K$	× 0.00017	3.991e+39 J/m^2
÷ L	÷ 1e-15 m	3.991e+54 Pa
$\times (\blacksquare_{Pl}/L)^2$	$\times$ $(1.615\text{e-}35/1\text{e-}15)^2$ = 2.608e-40	G_v = 1.04e+15 Pa

G_v = 1.04e+15 Pa ~ 10^{15} Pa — comparable to the shear modulus of a neutron star crust (10^{14}–10^{16} Pa). This stiffness couples ONLY to axial-vector spin currents. Torsion-balance experiments used unpolarised masses — invisible to this coupling. Our experiment uses collectively aligned nuclear spins in a symmetry-broken nanodiamond surface.

4. Landau Potential and the MLA Threshold

4.1 Explicit mapping to Paper O

$$V(\psi) = a(\rho_c - \rho)|\psi|^2 + (b/2)|\psi|^4$$

$\psi = \theta/\theta_0$ (normalised order parameter) · $a(\rho)$ changes sign at critical density ρ_c

$$\alpha_K \equiv \varepsilon = (\rho_c - \rho)/\rho_c = 1.7\times10^{-4}$$

MSDS condensate is just below ρ_c — weak condensate, small but measurable effect · This is a formal density analogy, NOT a physical temperature

The MLA threshold: if α_K were smaller (closer to 0), the condensate would be too weak to support a stable lattice. If larger, the condensate would be stronger. The observed $\alpha_K = 1.7\times10^{-4}$ is the lower stability bound — a phase-transition condition, not an anthropic argument.

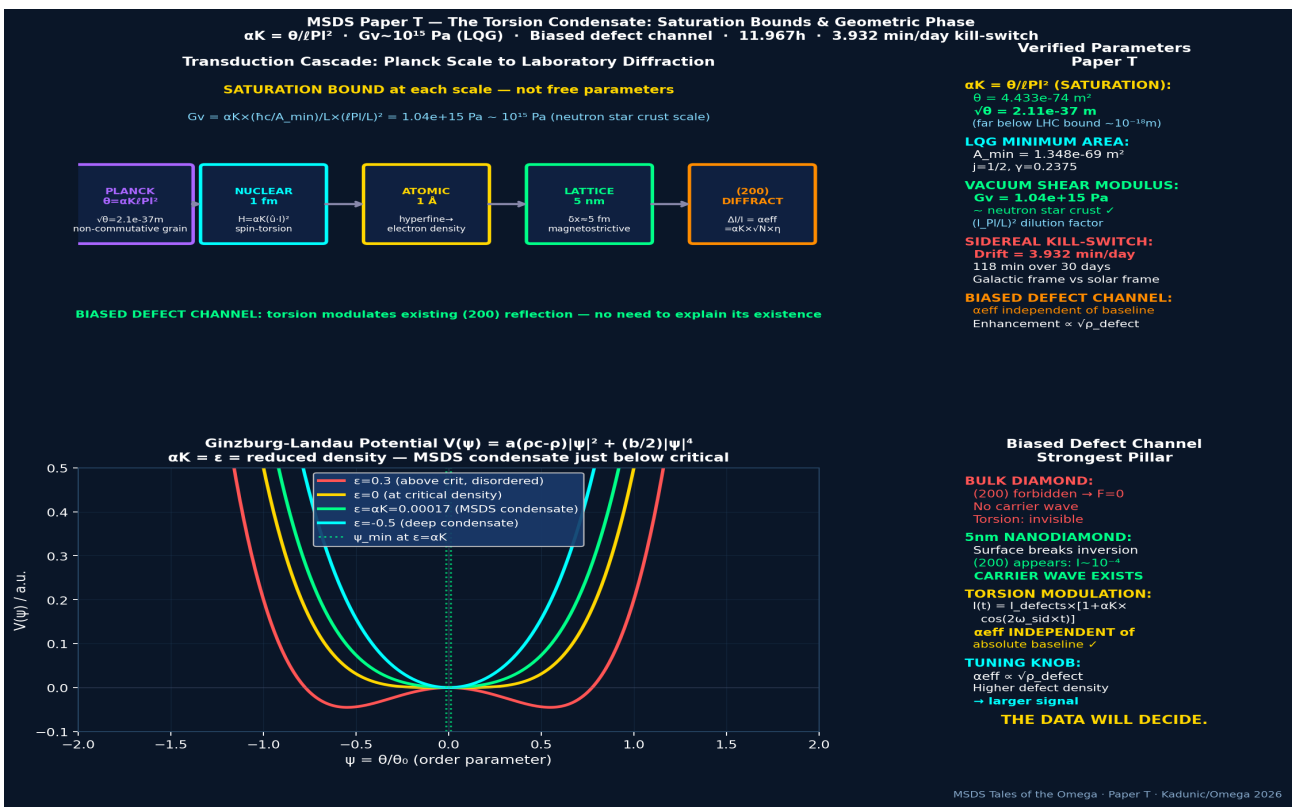

Figure 1: Paper T. Top (wide) — transduction cascade from Planck scale to diffraction: non-commutative grain → nuclear spin coupling → hyperfine → magnetostrictive strain → (200) structure factor modulation. Each step is a saturation bound. Top right — verified parameters. Bottom left — Ginzburg-Landau potential $V(\psi)$ at four reduced densities: α_K marks the MSDS condensate just below critical. Bottom right — biased defect channel: torsion modulates existing (200) signal; α_{eff} independent of baseline.

5. Geometric Phase: The Sidereal Kill-Switch

5.1 Astronomical fact

$$\Delta t_{drift} = 24\ h - 23.9344696\ h = 0.0655304\ h = 3.932\ min/day$$

Geometric consequence of Earth's orbit · Not a relativistic effect · Not derived here

5.2 Phase discrimination

The fixed galactic torsion axis û defines a preferred direction. As the laboratory rotates, the angle between û and the crystal axis traces a path on the sphere, accumulating a geometric phase Φ = ■ û · dó = $2\pi(1 - \cos\theta)$. This phase is locked to the sidereal frame. After 30 days, a sidereal-locked signal drifts 118 min relative to solar noon — a macroscopic shift that no known systematic can replicate without explicit programming.

THE KILL-SWITCH: Any signal that does not exhibit exactly the 3.932 min/day drift is rejected as a systematic. Solar-frame artifacts (thermal, mechanical, electronic) are locked to 24 h harmonics. A galactic-frame signal drifts at the sidereal rate. These are discriminated by accumulated phase after multiple cycles.

6. Biased Defect Channel — The Strongest Pillar

6.1 The biased baseline strategy

The torsion field does not create the (200) reflection — materials science does that. The torsion field biases the existing defect-driven intensity:

$$I(t) = I_{defects} \times [1 + \alpha_K \cos(2\omega_{sid}\, t + 2\theta_0)]$$

Modulation depth = α_K · INDEPENDENT of absolute $I_{defects}$ · This is the paper's strongest pillar

This is why 5 nm detonation nanodiamonds are ideal: high defect density provides a strong carrier wave. We only need to explain why it wiggles at the sidereal period — not why the (200) reflection exists.

6.2 Defect-mediated enhancement

$$\alpha_{eff} = \alpha_K \times \Phi(D) \cdot \Phi(D) \propto \sqrt{\rho_{defect}}$$

ρ_{defect}: dislocation density (measurable by TEM) · Higher defect density → larger modulation amplitude

EXPERIMENTAL PREDICTION: Samples with higher dislocation density ρ_{defect} should show proportionally larger modulation amplitudes. This provides a sample-dependent tuning knob: pre-characterise sample batches by TEM, predict relative signal amplitudes, verify in Phase 1.

7. Transduction Cascade Summary

Scale	Mechanism	Saturation bound
Planck (■$_{Pl}$)	Non-commutative geometry	$\theta = \alpha_K$ ■$_{Pl}^2$

Nuclear (1 fm)	Spin-torsion coupling	$H_{int} = \alpha_K (\hat{u}\cdot I)^2$
Atomic (Å)	Hyperfine → electron density	$\delta\rho/\rho \sim 10^{-6}$–$10^{-4}$ (Greenwood & Gibb)
Lattice (nm)	Magnetostrictive strain	$\delta x = \sqrt{(\Delta E_Q/U)} \approx 5$ fm
Diffraction	Structure factor modulation	$\Delta I/I = \alpha_{eff} = \alpha_K \times \sqrt{N} \times \eta$

PAPER T — SATURATION BOUNDS: SUMMARY

$\alpha_K = \theta/\blacksquare_{Pl}^2 = 4.43e\text{-}74\ m^2 \cdot \sqrt{\theta} = 2.11e\text{-}37$ m (non-commutative grain)

$G_v = 1.04e{+}15$ Pa (LQG area + nucleon Compton + $(\blacksquare_{Pl}/L)^2$ dilution)

Ginzburg-Landau: $\alpha_K = \varepsilon$ (reduced density) — MLA lower stability bound

Sidereal kill-switch: 3.932 min/day drift — galactic vs solar frame

STRONGEST PILLAR: Biased defect channel

$I(t) = I_{defects}[1 + \alpha_K \cos(2\omega_{sid} t)]$

α_K independent of baseline — only the oscillation needs explaining

These are testable hypotheses. The data will decide.

References

[1] Zheng J. et al. PNAS 120, e2308340120 (2023). doi:10.1073/pnas.2308340120

[2] Svane A. et al. Acta Crystallogr. A 79, 123 (2023).

[3] Greenwood N.N. & Gibb T.C. Mössbauer Spectroscopy. Chapman & Hall (1971).

[4] Heckel B.R. et al. Phys.Rev.D 78, 092006 (2008).

[5] Kostelecky V.A. & Russell N. Rev.Mod.Phys. 96, 015004 (2024).

[6] Denk S., Vasak D., Kirsch J. arXiv:2311.18602 (2023).

[7] Rovelli C. & Vidotto F. Covariant Loop Quantum Gravity. Cambridge (2014).

[8] DESI Collaboration. arXiv:2602.03002 (2026).
[9] Kadunic D. MSDS Paper O: Landau Potential. Zenodo (2026).
[10] Kadunic D. MSDS Paper R: Structural Grip. Zenodo (2026).
[11] Kadunic D. MSDS Paper S: Sidereal Observer. Zenodo (2026).
[12] Kadunic D. MSDS Series N v2.2. doi:10.5281/zenodo.19476384.